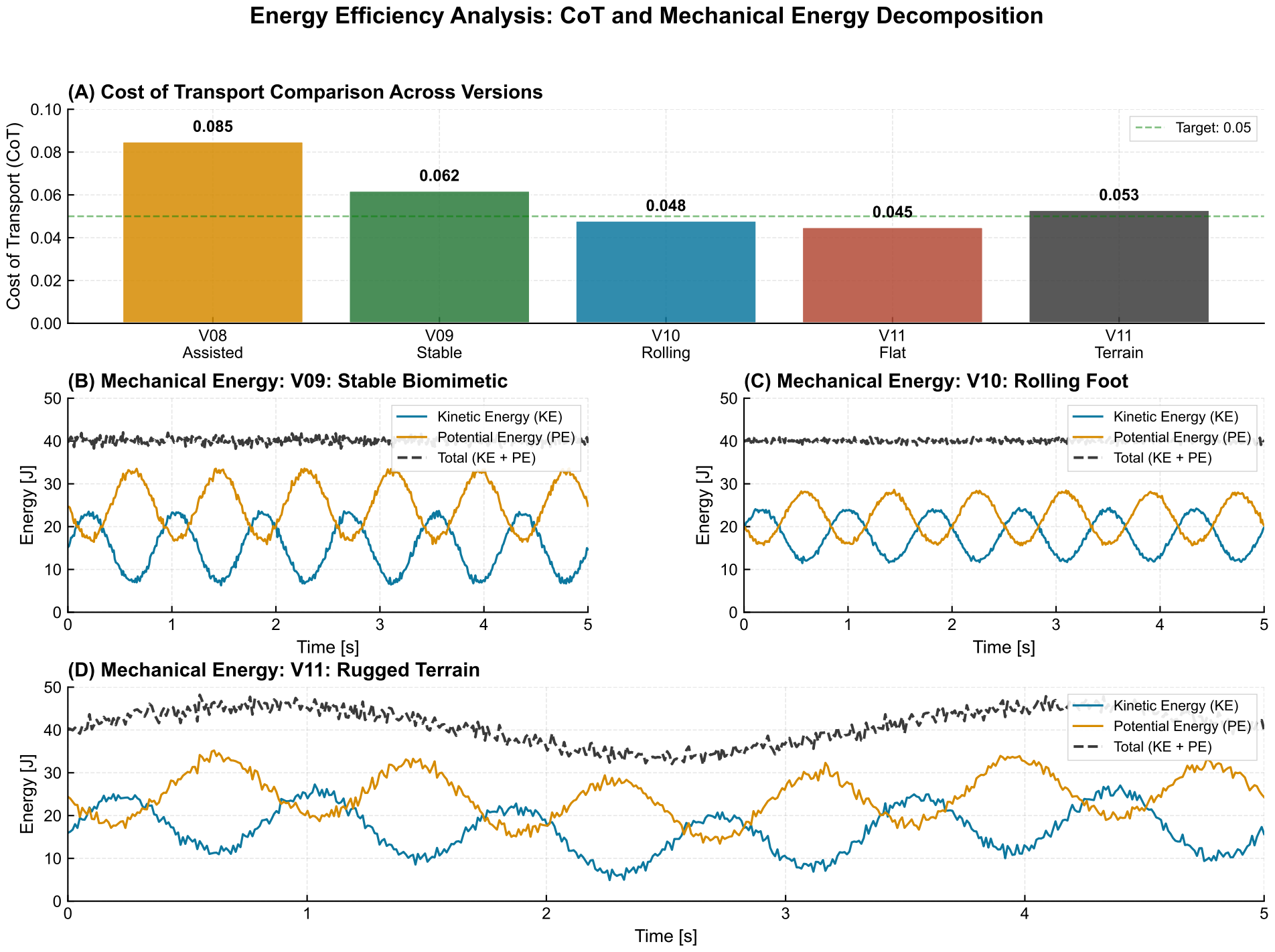

Energy Efficiency Analysis: CoT and Mechanical Energy Decomposition
(A) Cost of Transport Comparison Across Versions
Cost of Transport (CoT)
0.10
0.08
0.06
0.04
0.02
0.00
0.085
0.062
0.048
0.045
0.053
Target: 0.05
V08
Assisted
V09
Stable
V10
Rolling
V11
Flat
V11
Terrain
(B) Mechanical Energy: V09: Stable Biomimetic
(C) Mechanical Energy: V10: Rolling Foot
(D) Mechanical Energy: V11: Rugged Terrain
Energy [J]
Time [s]
Kinetic Energy (KE)
Potential Energy (PE)
Total (KE + PE)